Thomson Scattering

Sandro Vitenti

These notes use the conventions fixed earlier in the course, and build directly on the covariant radiation field derived in Radiation from a Moving Charge and the energy flux introduced in Energy of the Electromagnetic Field. The goal is to go from the covariant field of an accelerated charge, through its non-relativistic (local rest-frame) limit, to the classical differential and total cross sections for scattering of light by a free charge.

1 Setup

A free charge q of mass m (an electron, in practice) sits at rest at the origin and is illuminated by a monochromatic plane wave arriving from a distant source along the fixed direction $\hat{\mathbf{k}}$,

$$\mathbf{E}_{\rm inc}(\mathbf{x},t) = E_0\,\hat\epsilon\,\cos(\omega t - \mathbf{k}\cdot\mathbf{x}), \qquad \mathbf{k} \equiv \frac{\omega}{c}\hat{\mathbf{k}}, \qquad \hat\epsilon \perp \hat{\mathbf{k}},$$

which reduces, at the charge's location $\mathbf{x} = \mathbf{0}$, to $\mathbf{E}_{\rm inc}(t) = E_0\,\hat\epsilon\,\cos(\omega t)$ used below. This is a solution of the source-free Maxwell equations. Since $\hat\epsilon$ is a fixed vector, only the scalar factor is differentiated, with $\nabla\cos(\omega t - \mathbf{k}\cdot\mathbf{x}) = \mathbf{k}\,\sin(\omega t - \mathbf{k}\cdot\mathbf{x})$; Gauss's law $\nabla\cdot\mathbf{E}_{\rm inc} = 0$ then holds automatically,

$$\nabla\cdot\mathbf{E}_{\rm inc} = E_0\sin(\omega t - \mathbf{k}\cdot\mathbf{x})\,(\hat\epsilon\cdot\mathbf{k}) = 0,$$

by the transversality condition $\hat\epsilon \perp \hat{\mathbf{k}}$; and the wave equation $\Box\mathbf{E}_{\rm inc} = 0$ fixes $|\mathbf{k}| = \omega/c$, i.e. exactly the dispersion relation used above to define $\mathbf{k}$. The magnetic field follows from Faraday's law, $\nabla\times\mathbf{E}_{\rm inc} = -\partial_t\mathbf{B}_{\rm inc}$: using $\nabla\times(g(\mathbf{x})\hat\epsilon) = (\nabla g)\times\hat\epsilon$ for the same reason (any scalar function g times the fixed vector $\hat\epsilon$),

$$\nabla\times\mathbf{E}_{\rm inc} = E_0\sin(\omega t - \mathbf{k}\cdot\mathbf{x})\,(\mathbf{k}\times\hat\epsilon) = -\partial_t\mathbf{B}_{\rm inc},$$

which integrates in t (dropping an irrelevant static constant) to

$$\mathbf{B}_{\rm inc}(\mathbf{x},t) = \frac{E_0}{\omega}\cos(\omega t - \mathbf{k}\cdot\mathbf{x})\,(\mathbf{k}\times\hat\epsilon) = \frac{1}{c}\,\hat{\mathbf{k}}\times\mathbf{E}_{\rm inc}(\mathbf{x},t),$$

using $\mathbf{k} = (\omega/c)\hat{\mathbf{k}}$ in the last step. As a consistency check, this is exactly the radiation-field structure found in Radiation from a Moving Charge, $\mathbf{B}_{\rm rad} = \frac{1}{c}\hat{\mathbf{n}}\times\mathbf{E}_{\rm rad}$ with $\hat{\mathbf{n}}\cdot\mathbf{E}_{\rm rad} = 0$: a plane wave is simply what any radiation field looks like once its distant source is out of the picture, with $\hat{\mathbf{k}}$ now playing the role of the *incoming* propagation direction, in place of the outgoing $\hat{\mathbf{n}}$ used there to point from source to observer, and $\hat\epsilon \perp \hat{\mathbf{k}}$ the same transversality condition as $\hat{\mathbf{n}}\cdot\mathbf{E}_{\rm rad} = 0$. Either way, $|\mathbf{B}_{\rm inc}| = |\mathbf{E}_{\rm inc}|/c$, exactly as for any radiation field. Two approximations define the classical (Thomson) regime:

- **non-relativistic motion**: the charge's response velocity $v \ll c$, so terms of order v/c are dropped;
- **no radiation reaction**: the energy radiated per cycle is a small fraction of the charge's kinetic energy, so the incident field alone drives the motion.

2 Equation of motion

The charge's motion is governed by the covariant Lorentz force law,

$$m\,a^\mu = q\,F^{\mu\nu}u_\nu, \qquad a^\mu \equiv \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau},$$

with u^μ the four-velocity fixed in the conventions. This is the same content as the standard $\mathrm{d}p^\mu/\mathrm{d}\tau = qF^{\mu\nu}U_\nu$ with $p^\mu = mU^\mu$ and the textbook (dimensionful) $U^\mu = \mathrm{d}y^\mu/\mathrm{d}\tau = cu^\mu$; writing it in terms of this course's dimensionless $u^\mu = U^\mu/c$ multiplies both sides by c, which cancels, leaving the equation above with no explicit factor of c in front.

To extract the familiar 3-vector force law, specialize first to the instant at which the charge is at rest, $u^\mu = (1, \mathbf{0})$ — the same specialization used below for the radiation field — so that $u_0 = -1$, $u_i = 0$, and, exactly as there, $a^\mu = (0, \dot{\mathbf{v}}/c)$. Then

$$F^{i\nu}u_\nu = F^{i0}u_0 = -F^{i0} = F^{0i} = \frac{E^i_{\text{inc}}}{c},$$

with no contribution from the $F^{ij}u_j$ piece, since $u_j = 0$ at this instant: the magnetic force vanishes identically, not merely approximately, whenever $\mathbf{v} = \mathbf{0}$. The covariant equation of motion then reads $m\dot{v}^i/c = qE^i_{\text{inc}}/c$, i.e.

$$m\dot{\mathbf{v}} = q\mathbf{E}_{\text{inc}}(t).$$

This is exact only at the single instant used to define the rest frame; away from it $\mathbf{v} \neq \mathbf{0}$ and a magnetic force reappears. To see how small it is, repeat the computation without specializing u^μ, keeping the general lab-frame $u^\mu = \gamma(1, \mathbf{v}/c)$, so that $u_0 = -\gamma$, $u_j = \gamma v_j/c$. Then

$$F^{i\nu}u_\nu = F^{i0}u_0 + F^{ij}u_j = \frac{\gamma}{c}\Big[E^i_{\text{inc}} + (\mathbf{v} \times \mathbf{B}_{\text{inc}})^i\Big],$$

using $F^{i0} = -E^i_{\text{inc}}/c$ and $F^{ij}v_j = \epsilon^{ijk}B_k v_j = (\mathbf{v} \times \mathbf{B}_{\text{inc}})^i$, while $a^i = \mathrm{d}u^i/\mathrm{d}\tau = \gamma\,\mathrm{d}(\gamma v^i/c)/\mathrm{d}t$ using $\mathrm{d}/\mathrm{d}\tau = \gamma\,\mathrm{d}/\mathrm{d}t$. Substituting both into $ma^\mu = qF^{\mu\nu}u_\nu$ and cancelling the common factor γ/c from both sides gives the standard relativistic 3-force law,

$$m\frac{\mathrm{d}(\gamma\mathbf{v})}{\mathrm{d}t} = q(\mathbf{E}_{\text{inc}} + \mathbf{v} \times \mathbf{B}_{\text{inc}}).$$

Since $|\mathbf{B}_{\text{inc}}| = |\mathbf{E}_{\text{inc}}|/c$, the magnetic term here is smaller than the electric term by a factor v/c; in the non-relativistic regime $v \ll c$ assumed throughout (Setup) this holds at every instant along the trajectory, not just the one where the charge happens to be at rest, and $\gamma \approx 1$ to the same accuracy. Dropping both recovers exactly the rest-frame result found above, now understood to hold throughout the motion,

$$m\dot{\mathbf{v}} = q\mathbf{E}_{\text{inc}}(t), \qquad \dot{\mathbf{v}}(t) = \frac{q}{m}E_0\,\hat{\epsilon}\,\cos(\omega t).$$

The charge oscillates along the fixed direction $\hat{\epsilon}$ with acceleration amplitude $a_0 \equiv qE_0/m$; since $a_0/\omega \ll c$ in the non-relativistic regime, the charge's excursion is tiny compared to the wavelength, and it radiates essentially from a fixed point (the dipole approximation).

3 From the covariant field to the local frame

The radiation field of an accelerated charge, valid for any velocity, was written covariantly as

$$F^{\text{rad}}_{\mu\nu} = \frac{\mu_0 q}{4\pi(u \cdot \Delta x)^2}\left[\Delta x_\mu a_\nu - \Delta x_\nu a_\mu - \frac{\Delta x \cdot a}{u \cdot \Delta x}(\Delta x_\mu u_\nu - \Delta x_\nu u_\mu)\right]_{\tau=\tau_r},$$

with $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau$. Here this is specialized to a charge that is instantaneously at rest, as is the case for the non-relativistic oscillation driven by the incident wave.

To turn $F^{\text{rad}}_{\mu\nu}$ into an electric field, use the same covariant contraction as in the equation of motion above: the field measured by an observer of four-velocity w^μ is

$$E^\mu(w) \equiv c\,F^{\mu\nu}w_\nu,$$

automatically orthogonal to the observer, $w_\mu E^\mu(w) = 0$ (by antisymmetry of $F^{\mu\nu}$), and equal to $(0, \mathbf{E})$ in that observer's own rest frame: at $w^\mu = (1, \mathbf{0})$, $E^0(w) = cF^{00}w_0 = 0$ and, using $F^{i0} = -F^{0i} = -E^i/c$,

$$E^i(w) = cF^{i\nu}w_\nu = cF^{i0}w_0 = -cF^{i0} = cF^{0i} = E^i.$$

(The similar-looking contraction $u_\nu F^{\nu\mu} = -F^{\mu\nu}u_\nu$ has the opposite sign and is missing this factor of c; the correctly normalized object is $E^\mu = cF^{\mu\nu}u_\nu$, not $u_\nu F^{\nu\mu}$.)

Apply this with $w^\mu = u^\mu$, the radiating charge's own four-velocity, at the instant it is at rest: $u^\mu = (1, \mathbf{0})$ and, since $u_\mu a^\mu = 0$ forces $a^0 = 0$ there, $a^\mu = (0, \dot{\mathbf{v}}/c)$ exactly (this is the same identification used for the Larmor formula). With $\Delta x^\mu = (R, R\hat{\mathbf{n}})$, $R = |\mathbf{x}|$,

$$u \cdot \Delta x = -R, \qquad \Delta x \cdot a = R\,\frac{\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}}{c}.$$

Since $u^0 = 1$, $u^i = 0$, only the $\mu = i, \nu = 0$ component of the bracket in $F^{\text{rad}}_{\mu\nu}$ contributes to $E^{\text{rad}}_i = cF^{\text{rad}}_{i\nu}u^\nu = cF^{\text{rad}}_{i0}$. Using $a_0 = 0$, $u_0 = -1$, $u_i = 0$, $\Delta x_0 = -R$, $\Delta x_i = R\hat{n}_i$, and $\Delta x \cdot a/(u \cdot \Delta x) = -\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}/c$,

$$F^{\text{rad}}_{i0} = \frac{\mu_0 q}{4\pi R^2}\left[\Delta x_i a_0 - \Delta x_0 a_i - \frac{\Delta x \cdot a}{u \cdot \Delta x}(\Delta x_i u_0 - \Delta x_0 u_i)\right] = \frac{\mu_0 q}{4\pi R^2}\left[R\frac{\dot{v}_i}{c} + \frac{\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}}{c}(R\hat{n}_i)\right],$$

so that

$$E^{\text{rad}}_i = cF^{\text{rad}}_{i0} = \frac{\mu_0 q}{4\pi R}\Big[\dot{v}_i - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{n}_i\Big], \qquad \boxed{\mathbf{E}_{\text{rad}}(\mathbf{x}, t) = \frac{\mu_0 q}{4\pi R}\Big[\dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}}\Big]_{t_r}.}$$

The bracket is the component of $\dot{\mathbf{v}}$ transverse to $\hat{\mathbf{n}}$: writing $\dot{\mathbf{v}} = (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}} + \dot{\mathbf{v}}_\perp$, the bracket is exactly $\dot{\mathbf{v}}_\perp$, manifestly orthogonal to $\hat{\mathbf{n}}$. The same projection has an equivalent double-cross-product form: the BAC-CAB rule $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$ with $\mathbf{A} = \mathbf{B} = \hat{\mathbf{n}}$, $\mathbf{C} = \dot{\mathbf{v}}$, gives

$$\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}}) = \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}) - \dot{\mathbf{v}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}) = (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}} - \dot{\mathbf{v}},$$

using $\hat{\mathbf{n}} \cdot \hat{\mathbf{n}} = 1$, so $\dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}} = -\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}})$ and, equivalently,

$$\mathbf{E}_{\text{rad}}(\mathbf{x}, t) = -\frac{\mu_0 q}{4\pi R}\,\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}})\Big|_{t_r}.$$

Both forms make the transversality of the dipole radiation field to $\hat{\mathbf{n}}$ manifest: the first as an explicit orthogonal projection, the second because $\hat{\mathbf{n}} \times (\text{anything})$ is automatically perpendicular to $\hat{\mathbf{n}}$. Writing χ for the angle between $\hat{\mathbf{n}}$ and $\dot{\mathbf{v}}$, its magnitude is

$$|\mathbf{E}_{\text{rad}}| = \frac{\mu_0 q\,|\dot{\mathbf{v}}|}{4\pi R}\sin\chi.$$

4 Radiated power and a consistency check

In the radiation zone, $u = \varepsilon_0 E^2$ and $\mathbf{S} = cu\,\hat{\mathbf{n}}$ (from the Radiation zone section), so the power radiated per solid angle is

$$\frac{\mathrm{d}P}{\mathrm{d}\Omega} = R^2\,\mathbf{S} \cdot \hat{\mathbf{n}} = R^2 c\varepsilon_0|\mathbf{E}_{\text{rad}}|^2 = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c}\sin^2\chi,$$

using $1/(4\pi\varepsilon_0) = \mu_0 c^2/(4\pi)$. As a check, integrating over solid angle with $\int \sin^2\chi\,\mathrm{d}\Omega = 8\pi/3$ recovers exactly the non-relativistic Larmor formula already derived,

$$P = \int \frac{\mathrm{d}P}{\mathrm{d}\Omega}\,\mathrm{d}\Omega = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c} \cdot \frac{8\pi}{3} = \frac{\mu_0 q^2 \dot{v}^2}{6\pi c},$$

matching the boxed result $P = \mu_0 q^2 \dot{v}^2/(6\pi c)$ obtained independently from the covariant radiation tensor.

5 Differential cross section

The incident wave carries energy density $u_{\text{inc}} = \varepsilon_0 E_{\text{inc}}^2$ and flux $S_{\text{inc}} = c u_{\text{inc}}$ (same radiation-zone relations, since a plane wave is locally the same kind of transverse field); its time average over a cycle is

$$\langle S_{\text{inc}} \rangle = c\varepsilon_0 \langle E_{\text{inc}}^2 \rangle = \tfrac{1}{2} c\varepsilon_0 E_0^2.$$

What a cross section measures. A distant detector at angle (θ, ϕ) subtending solid angle $\mathrm{d}\Omega$ records scattered power $\langle \mathrm{d}P/\mathrm{d}\Omega \rangle \, \mathrm{d}\Omega$, independent of its distance (since $\mathrm{d}P/\mathrm{d}\Omega$ was built from the R-independent radiation field found above). The incident beam is instead characterized by its flux $\langle S_{\text{inc}} \rangle$, a power per unit *area* crossing a surface perpendicular to $\hat{\mathbf{k}}$. Dividing the two cancels the (arbitrary) incident intensity and turns "power per solid angle" into "area per solid angle" — hence the name *cross section*:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \equiv \frac{\langle \mathrm{d}P/\mathrm{d}\Omega \rangle}{\langle S_{\text{inc}} \rangle}, \qquad [\mathrm{d}\sigma/\mathrm{d}\Omega] = \text{area}.$$

Concretely, $\mathrm{d}\sigma$ is the cross-sectional area of the incident beam that carries exactly the power the charge re-radiates into $\mathrm{d}\Omega$; integrated over all directions (below), the total cross section σ is the area of a flat, perfectly absorbing target that would intercept, head-on, the same total power the charge actually scatters in every direction. It is the effective size the charge presents to the beam. Because both $\langle \mathrm{d}P/\mathrm{d}\Omega \rangle$ and $\langle S_{\text{inc}} \rangle$ scale as E_0^2, this ratio is independent of the incident intensity: it depends only on the scattering geometry and on q and m, which is what makes a cross section a property of the charge (or process) rather than of any particular beam. Using $\dot{\mathbf{v}} = (q/m)\mathbf{E}_{\text{inc}}$ from the equation of motion, $\langle \dot{v}^2 \rangle = (q/m)^2 \langle E_{\text{inc}}^2 \rangle = (q/m)^2 E_0^2/2$, so

$$\left\langle \frac{\mathrm{d}P}{\mathrm{d}\Omega} \right\rangle = \frac{\mu_0 q^2}{16\pi^2 c} \cdot \frac{q^2 E_0^2}{2m^2} \sin^2 \chi,$$

and dividing by $\langle S_{\text{inc}} \rangle = c\varepsilon_0 E_0^2/2$, with $1/\varepsilon_0 = \mu_0 c^2$,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\mu_0 q^2}{4\pi m} \right)^2 \sin^2 \chi.$$

Defining the **classical radius** of the charge,

$$r_q \equiv \frac{\mu_0 q^2}{4\pi m} = \frac{q^2}{4\pi\varepsilon_0 m c^2},$$

this is

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = r_q^2 \sin^2 \chi,}$$

with χ the angle between the observation direction $\hat{\mathbf{n}}$ and the (fixed) oscillation direction $\hat{\epsilon}$: scattering is strongest broadside to the oscillation and vanishes along it, exactly like a driven dipole antenna.

6 Unpolarized incident light

For unpolarized incident light, average $\sin^2 \chi$ over the two independent transverse polarizations. Take $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ and, without loss of generality, the observation direction in the x-z plane at scattering angle θ from $\hat{\mathbf{k}}$, $\hat{\mathbf{n}} = \sin\theta\, \hat{\mathbf{x}} + \cos\theta\, \hat{\mathbf{z}}$. For $\hat{\epsilon} = \hat{\mathbf{x}}$, $\sin^2 \chi = 1 - (\hat{\mathbf{n}} \cdot \hat{\mathbf{x}})^2 = \cos^2 \theta$; for $\hat{\epsilon} = \hat{\mathbf{y}}$, $\hat{\mathbf{n}} \cdot \hat{\mathbf{y}} = 0$ so $\sin^2 \chi = 1$. Averaging the two,

$$\langle \sin^2 \chi \rangle = \frac{\cos^2 \theta + 1}{2},$$

so that

$$\boxed{\left. \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \right|_{\text{unpol}} = \frac{r_q^2}{2} \left(1 + \cos^2 \theta \right),}$$

the classical Thomson formula, with θ now the angle between the scattered and incident directions.

7 Total cross section

Integrating over solid angle, with $x = \cos\theta$,

$$\sigma_T = \frac{r_q^2}{2}\int_0^{2\pi} \mathrm{d}\phi \int_{-1}^{1}(1+x^2)\,\mathrm{d}x = \pi r_q^2\left[x + \frac{x^3}{3}\right]_{-1}^{1} = \pi r_q^2 \cdot \frac{8}{3},$$

$$\boxed{\sigma_T = \frac{8\pi}{3} r_q^2.}$$

For an electron ($q = e$, $m = m_e$), r_q is the classical electron radius and $\sigma_T \approx 6.65 \times 10^{-29}\,\mathrm{m}^2$ is the Thomson cross section.

8 Validity

This is a purely classical, elastic calculation: the scattered light has the same frequency ω as the incident light, with no recoil. It holds provided:

- $v \ll c$ throughout the motion (already used to drop the magnetic force and to evaluate a^μ at $u^\mu = (1, \mathbf{0})$);
- the photon energy $\hbar\omega$ is much smaller than mc^2, so that the quantum recoil corrections of Compton scattering are negligible;
- the charge is genuinely free (or driven far from any resonance), so that no restoring force competes with the incident field in the equation of motion.

9 Exercises

9.1 Exercise 1: Dropping the Magnetic Force

a) Using $|\mathbf{B}_{\text{inc}}| = |\mathbf{E}_{\text{inc}}|/c$, estimate the ratio $|q\mathbf{v} \times \mathbf{B}_{\text{inc}}|/|q\mathbf{E}_{\text{inc}}|$ in terms of v/c, confirming that it is suppressed for non-relativistic motion.

b) Using $v \sim a_0/\omega$ for the oscillation velocity scale, with $a_0 = qE_0/m$, express v/c in terms of q, E_0, m, ω, c, and check that it is indeed small for visible light ($\omega \sim 10^{15}\,\mathrm{s}^{-1}$) incident on a free electron at ordinary laboratory intensities.

9.2 Exercise 2: The Local-Frame Radiation Field (Essential)

a) Starting from $E_i^{\text{rad}} = cF_{i\nu}^{\text{rad}}u^\nu$ specialized to $u^\mu = (1, \mathbf{0})$, with $\Delta x^\mu = (R, R\hat{\mathbf{n}})$ and $\Delta x \cdot a = R\,\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}/c$, write out F_{i0}^{rad} term by term and confirm

$$\mathbf{E}_{\text{rad}} = \frac{\mu_0 q}{4\pi R}\Big[\dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}}\Big].$$

b) Verify the equivalent form $-\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}}) = \dot{\mathbf{v}} - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})\hat{\mathbf{n}}$ using the BAC-CAB rule.

9.3 Exercise 3: Larmor Consistency Check (Essential)

a) Verify $\int \sin^2\chi\,\mathrm{d}\Omega = 8\pi/3$ by direct integration in spherical coordinates, with χ the polar angle measured from $\dot{\mathbf{v}}$.

b) Substitute this into $\mathrm{d}P/\mathrm{d}\Omega$ and confirm that the result reproduces exactly the Larmor formula $P = \mu_0 q^2 \dot{v}^2/(6\pi c)$ derived covariantly in Radiation from a Moving Charge, matching the numerical prefactor as well as the power of $\dot{v}$.

9.4 Exercise 4: The Differential Cross Section, Step by Step (Essential)

a) Using $\dot{\mathbf{v}} = (q/m)\mathbf{E}_{\rm inc}$ and $\langle E_{\rm inc}^2 \rangle = E_0^2/2$, verify $\langle \dot{v}^2 \rangle = (q/m)^2 E_0^2/2$ and substitute it into $\langle \mathrm{d}P/\mathrm{d}\Omega \rangle$ to reproduce the boxed intermediate expression given in the notes.

b) Divide by $\langle S_{\rm inc} \rangle$ and, using $1/\varepsilon_0 = \mu_0 c^2$, verify explicitly that every factor of c, μ_0, and m combines to leave $\mathrm{d}\sigma/\mathrm{d}\Omega = r_q^2 \sin^2\chi$, an area.

9.5 Exercise 5: Unpolarized Averaging in a General Direction

a) Starting from $\cos\chi = \hat{\mathbf{n}} \cdot \hat{\epsilon}$, verify $\sin^2\chi = 1 - (\hat{\mathbf{n}} \cdot \hat{\epsilon})^2$.

b) Using the rotational symmetry of the setup about $\hat{\mathbf{k}}$, argue that the unpolarized $\mathrm{d}\sigma/\mathrm{d}\Omega$ can only depend on the polar angle θ between $\hat{\mathbf{n}}$ and $\hat{\mathbf{k}}$, never on the azimuthal angle of $\hat{\mathbf{n}}$ — so it suffices to check the formula for $\hat{\mathbf{n}}$ in a single plane containing $\hat{\mathbf{k}}$, as the notes do.

9.6 Exercise 6: The Total Cross Section (Essential)

a) Verify $\int_0^{2\pi} \mathrm{d}\phi \int_{-1}^{1} (1 + x^2)\, \mathrm{d}x = 8\pi/3$ step by step, and hence confirm $\sigma_T = \frac{8\pi}{3} r_q^2$.

b) Using $r_q = q^2/(4\pi\varepsilon_0 m c^2)$ and $e = 1.6 \times 10^{-19}\,\mathrm{C}$, $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$, verify numerically that $\sigma_T \approx 6.65 \times 10^{-29}\,\mathrm{m}^2$ for an electron.

Feel free to create issues, ask questions, or suggest improvements in the GitHub repository.